Telescope Limits

Closure Criteria

Wind

Sustained 22mph or higher

Gusts above 30mph

Humidity

85% or higher

Opening Criteria

When there is sustained improvement for at least 20 minutes as indicated on website below (also linked to the 60 inch home page under "Ridge Weather")

Good rule of thumb: If Tierras is closed, the 60 inch should be as well.

Weather data available at:

https://veritas.sao.arizona.edu/~whanlon/ridgeWeather.html

FAST Afternoon and Morning Calibrations

Calibrations should start at least 3 hours prior to 12-degree twilight to ensure sufficient time to complete, deal with anomalies, and fill camera with LN2.

Dome should be closed and chamber lights off for all calibrations.

DARKS: "repeat 10 dark 900" (takes 10x15 min darks)

BIAS': "bias 20" (takes 20 bias images)

FLATs: In yellow NTCS GUI, click "Sky/Comp" so that it indicates "Comp". Click "Incand" so that it indicates "ON". When set: "repeat 20 flat 10"

BIAS' and FLATs are to be repeated in the morning once done with observing. DARKS only need to be done once; DARKS can be done in the morning but there's no guarantee that personnel won't disturb them by turning on a light.

DO NOT FORGET TO FILL CAMERA WITH LN2 AT THE END OF THE NIGHT. (Note: The Camera has a roughly 12 hours hold time and must be filled in the afternoon and morning).

60 Inch Startup and Shutdown

Startup:

1) Turn on drives on rack in Chart Room.

2) Verify that the Axis Motor Torque values are roughly "0" on the TCSGalil app on the computer in the control room. If one or both axes show a full deflection of a torque value, "bump" the corresponding axis button on the hand paddle to zero out the torque value. MAKE CERTAIN TORQUE VALUES ARE ZERO BEFORE TRYING TO SLEW TO AN OBJECT!

3) Turn "Dome On" and "Tracking On" under "Tele Tasks" drop down menu in yellow TCS GUI. If telescope is at zenith, the dome will now rotate to the east to track with telescope.

4) Verify the sun isn't a danger and open the slit via the following website: 192.33.141.174

5) Turn off chamber A/C via this website: 192.33.141.117

6) Open mirror covers via wall mounted switch in chamber.

Shutdown:

1) Using the "Controls" GUI on the Galil computer, click "Stow Mount at Zenith".

2) Verify the mount has stopped moving and turn off drives on rack in Chart Room.

3) Close mirror covers via switch in chamber

4) Close slit via 192.33.141.174 ; Turn on chamber A/C via 192.33.141.117

ALWAYS, ALWAYS, ALWAYS VERIFY TELESCOPES ARE STOWED AND DOME SLITS SHUT BEFORE DEPARTING AT THE END OF THE NIGHT

Getting Acquainted With the New 60" TCS ("Galil")

Introduction:

In 2019 the old "Harvey" telescope control system was replaced by the "Galil" control system manufactured by DFM.

The Harvey TCS was supported by a Windows '98 machine whereas the Galil is an "app" on a Windows 7 machine. The yellow NTCS GUI on the FLWO60 desktop remains identical to before, but some functionality has been lost, and there remains some compatibility issues.

The new Galil incorporates optical position sensors in RA and DEC as well as improved drive motors. As a result, not only is telescope pointing much improved, it is extremely rare for pointing to be lost.

The Galil makes use of three separate telescope position sensing systems, each with its own associated/enforced limits. The first limit typically encountered is sensed by a mercury switch mounted on the top southwest corner of the mirror cell. The second set of limits to be encountered are the mechanical "ramp" limits mounted on each axis. The third limit is sensed within the Galil itself. (There is a fourth "limit" to prevent the telescope from colliding with the eastern side of the north telescope mount. This limit is nothing more than a cable attached to a mechanical switch – the cable being intended to activate the switch upon impingement by the telescope.).

The old telescope "Rack" in the chart room is now home to the Galil computer (bottom of the rack) and telescope drive controls (top of the rack).

Start-up:

The Galil App will usually be running and visible on the desktop of the wall-mounted monitor in the control room. If it isn't, simply confirm that the guide computer is running (and in the appropriate configuration; guider up and running for FAST, no guider for Tress), the Realtime System is NOT running on FLWO60, and then double-click the icon labeled "Start TCS Galil" on the wall-mounted monitor under the window in the control room.

Once the Galil App is running, start the realtime system.

You are now able to go to the rack and turn the switch labeled "Drives" to on (all other switches typically remain on at all times).

Shut-down:

At the end of the night, in normal circumstances, the only thing to do is to turn the switch labeled "Drives" on the rack to off.

Useful GUIs Opened via the Galil App:

There are two useful GUIs opened through the "Operation" drop down menu on the main Galil Gui.

Controls GUI – This GUI is primarily useful for stowing the telescope at the end of the night.

Handpaddle GUI – This GUI is useful for focus moves, especially when Tres is mounted and you're using the automatic focus routine.

Galil Peculiarities:

As mentioned previously, there are a few compatibility issues between the Galil and the Realtime System.

The drop down menus on the yellow NTCS GUI still function. However, "Stow Tele" or "Stow All" in the yellow GUI will, more often than not, cause the yellow GUI to crash after things have stopped stowing.

The work-around is to stow the telescope through the Controls GUI opened via the Operation drop down menu in the Galil App: "**Stow Mount at Zenith**".

You can then return to the yellow GUI on FLWO60 to stow the dome.

Known Galil Issues:

The Galil occasionally goes dead in one or both telescope axes.

This usually happens at the start of the night, but may happen at any time while observing; you'll load new coordinates, click "Slew Enable" and discover that at least one axis has failed to move.

To recover,

Click "Stow Mount at Zenith" in the Controls GUI on the Galil monitor.
Exit the Realtime System on FLWO60.
Exit the Galil App (click the "x" on the upper right of the GUI).
Restart the Galil App.
Restart the Realtime System.
Repeat if necessary..

Recovering From a Limit Violation:

With the new Galil you can more often than not recover from a limit by simply using the hand paddle to manually back the telescope off the violated limit.

Should that not work, there are three "plugs" located in a ziplock bag on the rack to jumper the exceeded limit for whichever system that limit was violated. Since it isn't always easy to know which system flagged the limit, it is more time-effective to put all three plugs in at once. The plugs correspond to the electrical connectors on the back of the rack labeled, "**C2**", "**DIR LIMITS**", and "**INTERLOCK LIMITS**".

With those plugs in place, you can now move the telescope off the limit via the hand paddle.

How To Restart FAST and/or Galil for Whatever Reason Including an Axis Failure

If, after turning on the telescope drives, one or both of the axis torque indicators (RA and DEC) is pegged at a limit, the telescope will be inoperable until the torque indicators have been zeroed out.

Usually, momentarily pressing the axis button opposite the torque indicator's direction on the hand paddle will accomplish this. However, if this fails or this step is missed, recover as follows.

Exit the Realtime System on FLWO60 by typing "bye" or "exit" in the black "Telshell" window (you may also bring up a new xterm and type "killcom" if for some reason there is no command line in the Telshell).

Once all windows affiliated with the Realtime System are gone (IRAF and its windows will still remain), exit the TCS App on GALIL by clicking the "X" in the upper right corner of the main GUI (the GUI containing the torque and telescope positional indicators) and clicking "yes" in the pop-up window which follows. (Note: "GALIL" and "TCS App" are used interchangeably. However, GALIL is the name of the computer on which the TCS App is found).

Once the TCS App and its windows are gone, double click on the yellow "TCS Galil" app on the desktop. The app should promptly start and be operational again.

Next, it is necessary to confirm that the previous "tcsserver" has been stopped on FLWO60. In an xterm window on FLWO60, type:

```
ps aux | grep tcsserver
```

Look to see if there is still a tcsserver running, if so type:

```
kill -9 *process number*
```

You should now be able to restart FAST (aka, "Realtime System") by typing "gofast &" in any xterm window on FLWO60.

(NOTE: If you are unable to kill the previous tcsserver, it will eventually time out in 5 to 10 minutes.)

FAST IRAF Edit Parameters (epar qfast)

PACKAGE = procdata
TASK = qfast

Nobs =	90	FAST image number to process
(roodir =	/rdata/fast/)	Spectrum image root directory
(obsdate =	2026.0218)	Date of observation directory (yyyy.mmdd)
(fcal =	comp300+600)	Calibration file name
(caldir =	qfast)	Calibration image directory
(row1 =	5)	First row for spectrum search
(row2 =	155)	Last row for spectrum search
(keepnam =	yes)	Keep object name in file name (y or n)
(apdisp =	yes)	Display aperture and background (y or n)
(imdisp =	no)	Display raw image (y or n)
(spdisp =	yes)	Display spectrum (y or n)
(quickre =	no)	Cross-correlate redshift (y or n)
(quickem =	no)	Emission line redshift (y or n)
(delquic =	no)	Delete quick-look 1-D spectrum (y or n)
(verbose =	no)	Log progress (y or n)
(mode =	h)	

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60" Observer Collateral Duties

Oversee operation of 48" Telescope and robot.

Overview:
The 48" telescope utilizes a "robot" and script to prioritize and select objects to observe throughout the night. Under normal circumstances its operation is imperceptible to the 60" observer. However, it's necessary to keep an eye on things.

Monitoring 48" Images:
VNC into the 48" desktop-

Open an xterm on the 60" desktop and type the command, "vncviewer flwo48:2 &" (password is available on the 60" monitor).

Open up a DS9 on 48" desktop-

In a xterm window type "ds9 &". However, there will likely be another ds9 window open on the third desktop on flwo48.

Pull up the latest 48" images on the ds9-

On the ds9 GUI, click "File" > "Open Other" > "Open Mosaic IRAF". At the bottom of the new pop-up, change the directory to the current file, ie: "/rdata/kep/YYYY.MMDD", and click return. You can now select the desired image from the newly populated "files" list in the "Open" gui.

Controlling the 48" Open/Close Status When the Weather is Sketchy:
The 48" weather sensor system is normally good at detecting weather changes. However, there are certain cloud conditions that the sensor has difficulty with, particularly when the coloring of the clouds is relatively uniform. There also instances when ambient conditions simply aren't producing worthwhile data.

VNC into the 48" and go to the 2nd desktop-

On the 2nd desktop you'll find the GUI labeled "FLWO 1.2m Weather". Within this GUI is an entry field labeled "sky temperature". After clicking on this entry field, you can adjust the open/close temperature criteria. Normal temperature range is between -29 and -36. However, if conditions simply aren't allowing for worthwhile data, you can adjust the temperature as low as needed to force the dome to close and the 48" to shut down.

You can also reverse the above to force the telescope to open.

Note that there are several monitored criteria. Humidity and Wind should be left as-is. But night closings can be increased if necessary and clicking on either "Weather Timeout" or "Sky Timeout" will bypass the countdown remaining allowing the dome to open immediately.

48" Error Recovery:
Should the 48" encounter an error, you will receive notification at the 60". Most notifications will give a brief description of the error. Because of the variety of errors, it's impossible to cover each specific case.

However, most all errors can be corrected by simply exiting the robot and using the hand-paddle or other corrective actions.

To Stop The 48" Robot-

Type "cntrl-C" in the 48" Telshell (or "STOProbot" in an xterm window). Now, in an xterm window type, "Rstop". This will exit the robot and allow manual control of the telescope to facilitate recovery from the error. Once the error has been corrected, type "clear_robot" to stop continued notification of the error in the 60" control room.